

**Department of Electronics and Communication Engineering****Academic Year 2022 – 2023 (Even Semester)****Degree, Semester & Branch: IV Semester B.E. ECE B****Course Code & Title: EC3492 Digital Signal Processing****Name of the Faculty member (s): Mrs.V.SrIREngaNachiyar****Innovative Practice Description**

- **Unit / Topic:** Unit V / Fixed point architecture principles
- **Course Outcome:** CO 5
- **Unit Outcome:** 5c
- **Activity Chosen:** Flipped class
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of "Fixed point architecture principles". They described the DSP architecture. Students were divided into groups and each group will prepared the detailed presentation of Fixed point architecture principles and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO3
CO5	2	2	3	1	2	2	1	2

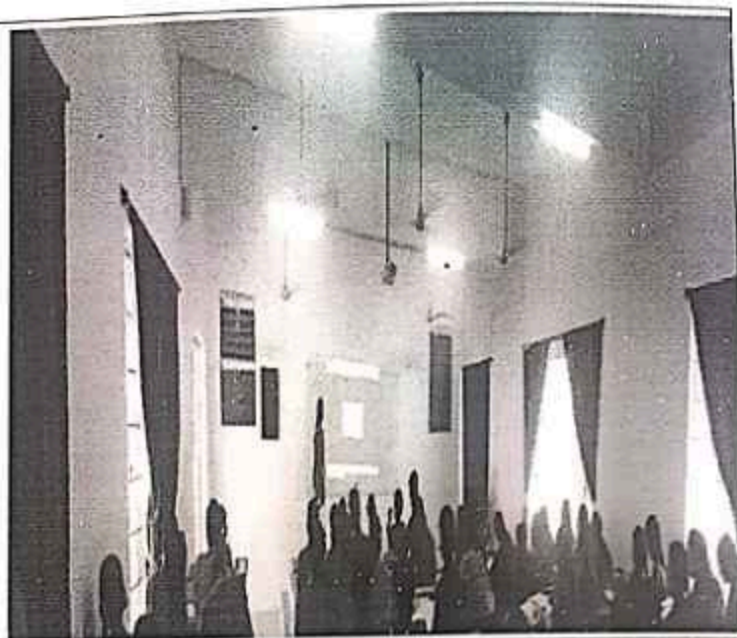
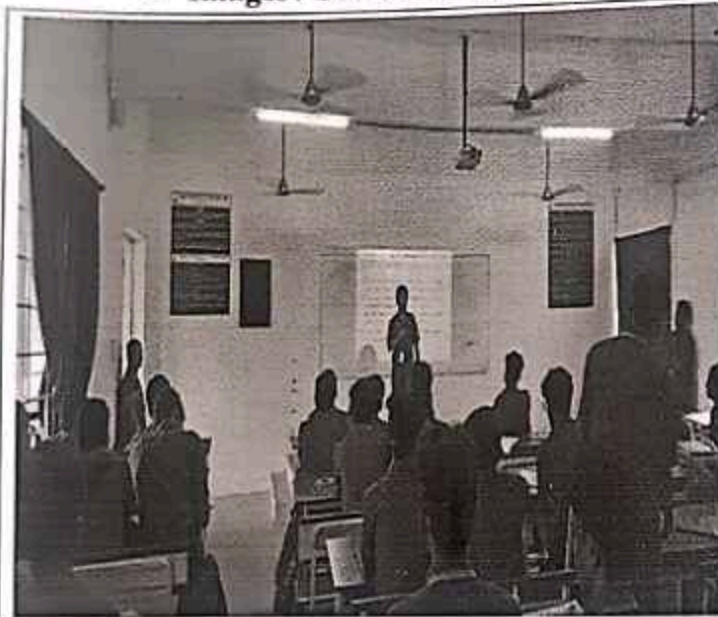
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PO12
	2	1
Justification for correlation	Students will understand the DSP Processors architecture and will give clear information during discussion session. Hence it is mapped at level 2	Students were identify the previous instances in architecture where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus level 1 is mapped



• Images / Screenshot of the practice:



Reflective Critique:

❖ **Feedback of practice from students and other stakeholders:**

The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge of various DSP processors. Due to this activity, the students were able to recall the topic and remember the important terms easily.

❖ **Challenges faced in implementation:**

I have planned this activity for 30 minutes. Students take some time to present the topic. After formed the group, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the various DSP processors and encouraged the students to present which it takes another 5 minutes for completing the activity.

References:

- ❖ Digital Signal Processing – Principles, Algorithms & Applications, John G. Proakis & Dimitris G. Manolakis, Pearson Education / Prentice Hall, 2007 ISBN No. : 0-13-394338-9.
- ❖ Discrete-Time Signal Processing A. V. Oppenheim, R.W. Schaffer and J.R. Buck Pearson, 2004 ISBN No. : 0-13-754920-2.

Signature of the faculty member



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Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. ECE B

Course Code & Title: EC3492 Digital Signal Processing

Name of the Faculty member (s): Mrs.V.SrIREngaNachiyar

Innovative Practice Description

- **Unit / Topic:** Unit IV / Coefficient quantization error
- **Course Outcome:** CO 4
- **Unit Outcome:** 4b
- **Activity Chosen:** Jigsaw
- **Justification:**

Jigsaw is a cooperative group activity in which students are interdependent to achieve a common goal. In part one, each group is provided a different problems in Input /Output Quantization Errors. The group members become experts on that particular topic and create a group response.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

Students said that they felt this activity had improved their understanding of the Coefficient quantization error. Because in this unit, few problems were solved in same method. Some changes will be implemented for the particular topic. During assessment tests, it is simple to recall the Coefficient quantization error.

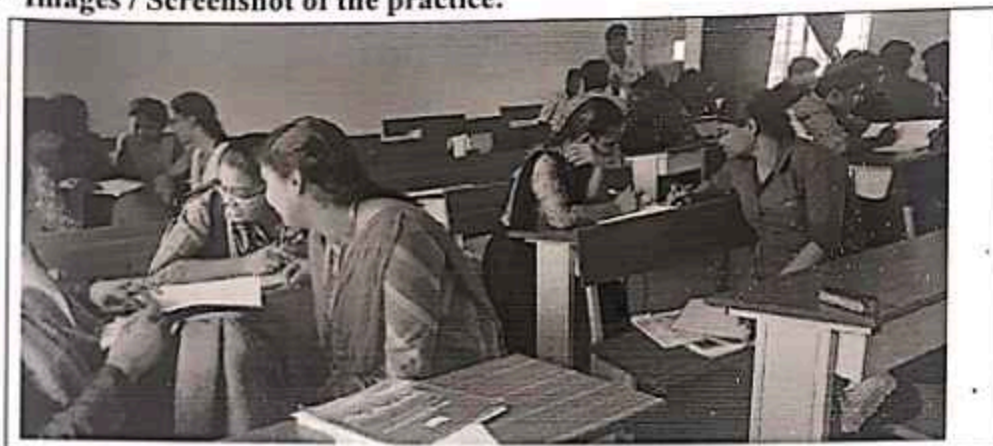
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO3
CO4	1	2	2	2	3	2	1
(1 – Low 2 – Moderate 3 – High)							

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Students discussed with their peers and write the procedures and solving techniques in paper. So, it is mapped at level 3

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students stated that they felt that this activity helped them to understand the procedure and steps to calculate the coefficient quantization error. It is very useful to recollect the important steps for solving

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The Students eagerly participated in that activity and then they acquired the knowledge about various error occurred while transmitting a digital signal. Due to this activity, the students were able to recall the topic and remember the important terms easily.

❖ **Challenges faced in implementation:**

Students took some time to forming the group. After formed the group, they discussed the topic and present the important terms in paper. I encouraged the students for completing the activity with useful points.

References:

- ❖ Digital Signal Processing – Principles, Algorithms & Applications, John G. Proakis & Dimitris G. Manolakis, Pearson Education / Prentice Hall, 2007 ISBN No. : 0-13-394338-9.
- ❖ Discrete-Time Signal Processing A. V. Oppenheim, R.W. Schaffer and J.R. Buck Pearson, 2004 ISBN No. : 0-13-754920-2.

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Department of Electronics and Communication Engineering

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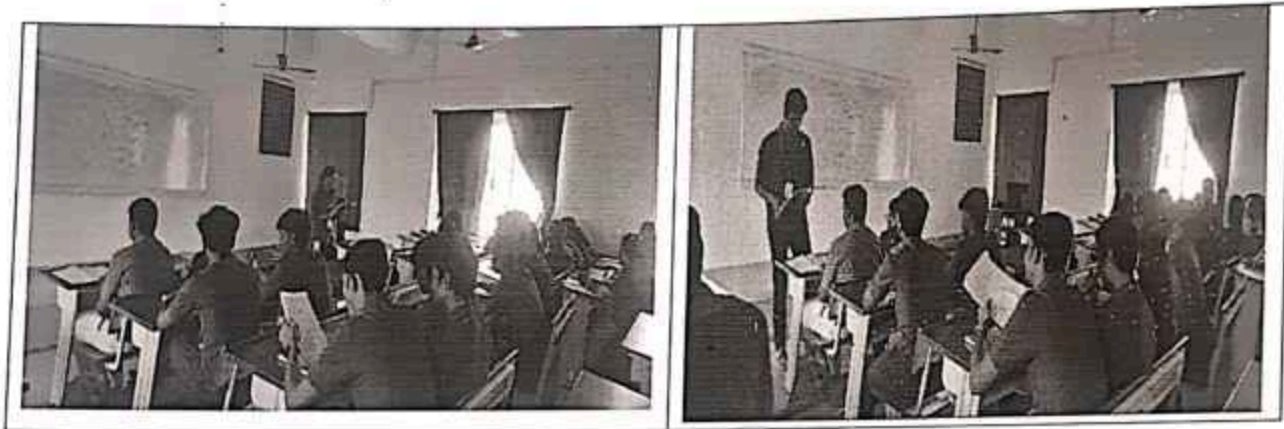
Name of the Faculty member: Mrs. V.SirengaNachiyar

Course Code & Title : EC3492 Digital Signal Processing

Semester, Degree & Branch : IV B.E ECE B

Topic : FIR filter structures - direct form realizations

Innovative Method : Zero Minute Speech



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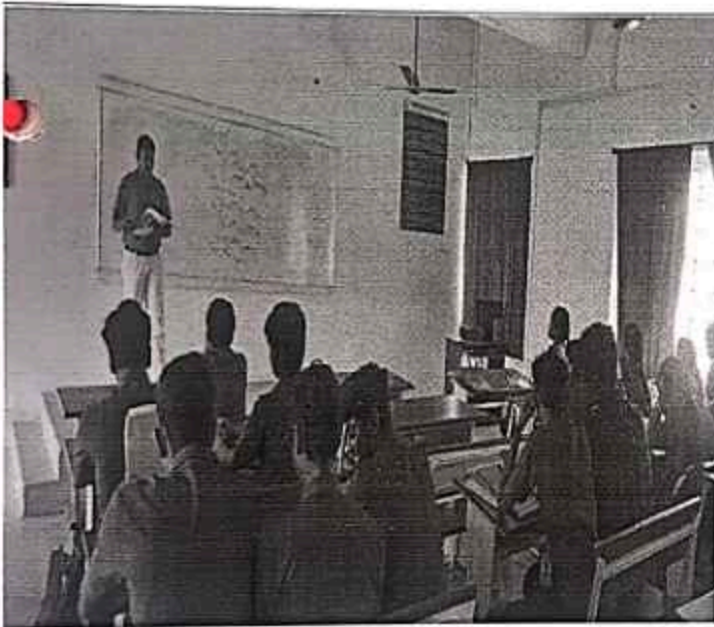


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Department of Electronics and Communication Engineering
Academic Year: 2022 - 2023 (Even Semester)

Name of the Faculty member: Mrs.V.SrirenaNachiyar
Course Code & Title : EC3492 Digital Signal Processing
Semester, Degree & Branch : IV B.E ECE B
Topic : Cascade, parallel realization
Innovative Method : Brainstorming



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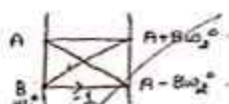
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Department of Electronics and Communication Engineering
Academic Year: 2022 - 2023 (Even Semester)

Name of the Faculty member: Mrs. V. Sirenga Nachiyar
Course Code & Title : EC3492 Digital Signal Processing
Semester, Degree & Branch : IV B.E ECE B
Topic : Linear filtering using FFT
Innovative Method : One Minute Paper

2A-point DIT-FFT Butterfly diagram



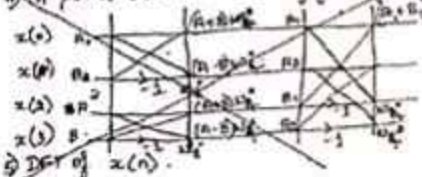
2A-point DIT FFT Bit-reversed order

$$\begin{aligned} x(0) &= x(0) \\ x(1) &= x(2) \\ x(2) &= x(1) \\ x(3) &= x(0) \end{aligned}$$

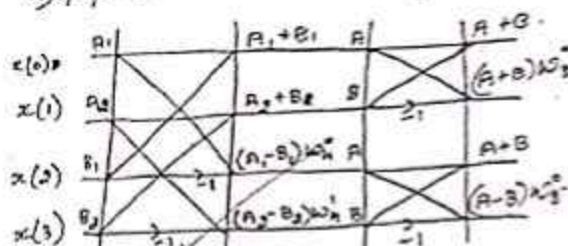
Twiddle factor for 8-point-FFT

$$\begin{aligned} W_8^0 &= 1 & W_8^2 &= -j \\ W_8^1 &= 0.707 - j0.707j & W_8^3 &= -0.707 - j0.707j \end{aligned}$$

4-point DIF-FFT Butterfly diagram



4-point DIF-FFT Butterfly diagram



DFT of $x(n)$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

$$k = 0, 1, 2, \dots, N-1$$

16-point DIT-FFT butterfly diagram



$$\begin{aligned} X(0) &= x(0) + x(8) + x(4) + x(12) \\ X(1) &= x(0) - x(8) + j(x(4) - x(12)) \end{aligned}$$

$$\begin{aligned} W_8^0 &= 1 \\ W_8^1 &= 0.707 - j0.707j \\ W_8^2 &= -0.707 - j0.707j \\ W_8^3 &= -0.707 + j0.707j \end{aligned}$$



16-point DIT-FFT butterfly diagram

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

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